

Cerevia Migraine Hackathon Development Guide

Standalone migraine-focused healthtech demo: website, GP dashboard, iOS patient app, Supabase backend and AI-powered consultation intelligence

This guide replaces the previous general chronic-condition version with a standalone migraine-focused product. Do not mention any other product or brand in the demo, pitch, codebase, website copy, app copy, repository name or generated outputs.

Master Development Prompt for Cerevia

Copy and paste this prompt into the development agent first. A separate branding prompt should follow afterwards. The end of this master prompt explicitly tells the agent not to finalise branding until those guidelines are provided.

Copy-Paste Master Prompt

You are an expert full-stack product engineer, iOS engineer, healthtech UX engineer, AI engineer, data visualisation engineer and technical architect.

You are helping build Cerevia for a hackathon. Treat this as a serious startup-grade demo, not a throwaway migraine diary app.

Product context:

Cerevia is an AI-powered migraine management and consultation-intelligence platform. It helps migraine patients capture detailed migraine episodes, triggers, medication usage, wearable context and daily impact between GP appointments. It then transforms that history into structured, visual, GP-ready insight so a GP can understand months of migraine progression in minutes rather than relying on patient memory alone.

Core problem:

Migraine patients often experience attacks, aura, pain-location changes, medication effects, triggers and lifestyle impacts between appointments. During a short GP consultation, they may struggle to remember frequency, severity, duration, medication effectiveness, suspected triggers and impact on work or study. GPs need to know whether the patient is improving, worsening, responding to treatment, or needs further review, but the available history is often incomplete.

Core value proposition:

Cerevia helps patients and GPs communicate more effectively by turning migraine history into structured events, visual trends, a 3D Migraine Twin, trigger context, medication-effectiveness views and concise consultation summaries.

Primary clinical user:

The clinician user is a GP. The GP dashboard must help answer four questions quickly:

1. Has the patient's migraine pattern improved or worsened?
2. Has migraine frequency, severity or duration changed since the last appointment?
3. Does the current medication appear to be helping based on patient-reported data?
4. What should the GP discuss during the consultation?

Product safety:

Cerevia must never diagnose, prescribe, recommend medication changes, provide treatment advice or replace clinical judgement. It should use safe wording such as "reported", "appears", "may be worth discussing", "patient-reported", "possible pattern" and "suggested consultation focus". It must include a clear disclaimer that it organises patient-reported migraine information and does not provide medical advice.

Final hackathon output:

1. A public-facing website called "Cerevia Hackathon Demo".
2. The website landing page must offer two clear options: "Continue as Patient" and "Continue as GP".

3. The patient experience should be a native-style iOS app that can be opened and viewed through Xcode. Prefer SwiftUI unless there is a strong reason to use another approach.
4. The GP experience should live on the website as a desktop-first GP dashboard.
5. Supabase must be used as the shared backend for the website, GP dashboard and patient app.
6. Use one deterministic seeded demo patient for speed and reliability.
7. The Migraine Twin must appear in both the patient app and GP dashboard.
8. Triggers must be a major part of the product, not a side note.
9. Apple Watch data should be simulated for the hackathon and include sleep, activity and heart-rate context.
10. The architecture should be realistic enough to discuss with technical judges, clinicians and startup mentors.

Primary demo patient:

Use one deterministic seeded patient so every demo run is predictable.

Recommended patient:

- Name: Sarah Ahmed
- Age: 29
- Primary language: English
- Condition: Chronic migraine
- Current medication: Sumatriptan, plus simple pain relief as patient-reported context if needed
- Recent pattern: migraine frequency increasing over the last 8-12 weeks
- Main migraine features: unilateral temple pain, behind-eye pain, aura, light sensitivity, nausea, dizziness, neck tension
- Triggers/context: poor sleep, high stress, screen time, skipped meals, dehydration, weather change and menstrual-cycle context
- Wearable context: reduced sleep, lower activity on migraine days, elevated resting heart rate around severe episodes
- Appointment context: upcoming GP appointment and previous appointment comparison
- Patient concern: migraines are becoming more frequent and affecting work productivity

Core product loop:

Patient opens the iOS app -> logs a detailed migraine episode -> selects pain location on a Migraine Twin -> records triggers, medication and impact -> optional voice/typed note is structured by AI -> Supabase stores the Migraine Event -> patient timeline updates -> Migraine Twin updates -> patient shares access with GP -> GP opens the website dashboard -> GP sees frequency, severity, duration, treatment effectiveness, trigger context, timeline replay, Migraine Twin and AI consultation summary.

Website and GP dashboard architecture:

- Recommended stack: Next.js App Router, TypeScript, Tailwind CSS, shadcn/ui, Recharts, Framer Motion, Supabase JS client, React Three Fiber or Three.js for the GP Migraine Twin.
- The landing page must say "Cerevia Hackathon Demo" and clearly route to Patient or GP.
- GP dashboard must include patient overview, key migraine changes, migraine frequency trend, severity/duration trend, medication effectiveness, trigger context, migraine event timeline, Migraine Twin, Consultation Copilot, Ask Cerevia, previous appointment comparison, wearable context and generated report.

iOS patient app architecture:

- Recommended stack: SwiftUI, Supabase Swift client, local state where needed, SceneKit or RealityKit for a simple 3D Migraine Twin.
- The app should feel like a real patient mobile product, not a website wrapped in a phone.
- Screens should include dashboard, migraine episode logging, trigger logging, medication/effectiveness tracking, voice or typed note input, migraine timeline, Migraine Twin, share with GP and appointment preparation.
- The app must write to Supabase so the GP dashboard can reflect patient actions.

Supabase backend:

Use Supabase for PostgreSQL, Auth, Storage, Row Level Security, reports, voice note metadata and deterministic seed data. Build a clear relational schema, but keep it hackathon-realistic.

Core tables to create:

- profiles
- patients
- gp_profiles
- medications
- migraine_events
- migraine_symptoms
- migraine_triggers
- medication_logs
- voice_notes
- gp_access
- reports
- appointments
- wearable_events

Migraine Event architecture:

The Migraine Event model is the core data foundation. A migraine episode is not just a note. It should capture: timestamp, severity, duration, pain region, aura, associated symptoms, suspected triggers, medication taken, time to relief, impact and linked wearable context.

Event types should include:

- migraine_episode_logged
- trigger_logged
- medication_taken
- medication_effectiveness_logged
- voice_note_added
- wearable_data_synced
- gp_access_granted
- report_generated
- appointment_prepared
- previous_appointment_compared

AI layer:

Mock AI first with realistic hardcoded outputs. Wire live AI later only after the end-to-end flow works.

AI functions:

1. Migraine consultation summary - summarise migraine history for a GP.
2. Voice note structuring - turn a voice transcript into structured migraine episode fields.
3. Ask Cerevia - let the GP ask questions about migraine history.
4. Pattern detection - highlight trends and correlations without diagnosis.
5. Previous appointment comparison - compare current migraine state to previous consultation.
6. Medication effectiveness summary - summarise patient-reported medication response.
7. Trigger context summary - show which triggers are commonly reported before severe episodes.

Migraine Twin:

The Migraine Twin is not a medically accurate diagnostic model. It is a visual representation of patient-reported migraine pain regions and symptom patterns. Use a simplified 3D head model or clean head silhouette with regions such as left temple, right temple, forehead, behind left eye, behind right eye, back of head and neck. It should show severity, frequency, recency and progression over time.

Wearables:

For the hackathon, use simulated Apple Watch data. Do not build real HealthKit integration unless there is enough time. Simulated data should include sleep, activity and heart rate. Wearable data should become wearable events and provide context, not diagnosis.

Build priority:

- Phase 1 - Set up Supabase, website, iOS app and seed data.
- Phase 2 - Build core migraine logging and GP dashboard.
- Phase 3 - Implement migraine events, timeline and sharing.

- Phase 4 - Add AI mock outputs, then optional live AI.
- Phase 5 - Add Migraine Twin in both experiences.
- Phase 6 - Add Consultation Copilot, Ask Cerevia and previous appointment comparison.
- Phase 7 - Add simulated Apple Watch data, trigger insights and timeline replay.
- Phase 8 - Polish the demo story and test the full flow.

Acceptance criteria:

- The website landing page clearly says "Cerevia Hackathon Demo".
- Users can choose Patient or GP from the landing page.
- The patient app opens in Xcode/iOS simulator and shows Sarah's migraine dashboard.
- Patient can log an advanced migraine episode.
- Patient actions create Migraine Events in Supabase.
- The GP dashboard reads Sarah's data from Supabase.
- The Migraine Twin appears in both patient and GP experiences.
- The GP can generate or view a migraine consultation summary.
- The GP can ask at least three useful questions through Ask Cerevia.
- Trigger context and medication effectiveness are visible on the GP dashboard.
- The system includes clear safety boundaries around AI.
- The demo is deterministic and does not break if the internet/API fails.

Important implementation instruction:

Before building the final UI styling, wait for the follow-up branding prompt. Do not finalise colours, typography, illustration style, icon style, spacing scale, shadows, animations or product copy tone until the branding guidelines are provided. Use neutral placeholder styling where necessary, but keep the architecture clean so branding can be applied after this prompt.

Step-by-Step Development and Idea Guide

This section expands the master prompt into a working development plan for the team. It explains the product idea, the architecture, the build order, the demo flow and the safety boundaries. It is written for a team building before and during a hackathon.

1. What Cerevia Is

Cerevia is a migraine-focused healthtech demo designed to help patients record what happens between GP appointments and help GPs understand migraine history quickly. The product is not a generic symptom tracker. It is a migraine consultation-intelligence platform.

Cerevia captures migraine episodes, pain location, aura, associated symptoms, suspected triggers, medication use, treatment response, wearable context and daily impact. It then turns that information into a timeline, visual trends, a Migraine Twin, Ask Cerevia answers and a GP-ready consultation summary.

Core positioning: Cerevia turns months of migraine history into structured, visual, GP-ready insight so clinicians can understand progression in minutes rather than relying on patient memory alone.

2. Why It Matters

- Migraine patients may remember the most recent attack, but struggle to summarise the last 8-12 weeks accurately.
- GPs need frequency, severity, duration, medication response and impact, not just a short description of pain.
- Migraine symptoms can include aura, nausea, light sensitivity, sound sensitivity, dizziness and neck tension, which are often poorly structured in standard notes.
- Triggers such as sleep, stress, hydration, screen time, missed meals, weather and menstrual-cycle context can be relevant for discussion, but should not be treated as diagnosis.
- Medication effectiveness matters: did it help, how quickly, and did the pattern change over time?
- Wearable data can provide context around sleep, activity and heart rate, but must remain supportive information rather than clinical proof.
- A migraine-specific visual twin can help patients and GPs understand pain location patterns at a glance.

3. Final Hackathon Output

Component	What it is	Purpose
Public demo website	Landing page with Patient and GP choices	Gives judges a clear entry point into the demo
Patient iOS app	Native-style SwiftUI app opened through Xcode/iOS simulator	Shows the patient experience of logging migraines, triggers, medication and impact
GP dashboard	Desktop dashboard hosted on the website	Shows GP-facing migraine insights, summary, trends and Migraine Twin
Supabase backend	Shared database, auth-lite, storage and access logic	Connects patient app and GP dashboard
Seeded patient	One deterministic chronic migraine patient profile	Keeps the demo controlled and reliable
Migraine Twin	3D head model or head-region visual in both experiences	Makes migraine pain location and progression understandable at a glance
AI layer	Mock-first intelligence functions	Summaries, Ask Cerevia, voice structuring and comparison without diagnosis

4. Demo Architecture Overview

Website landing page

```
|-- Continue as Patient -> explains/launches iOS patient app demo
|-- Continue as GP -> opens GP dashboard on website
```

Patient iOS app through Xcode

```
|-- Dashboard
|-- Migraine episode logging
|-- Trigger logging
```

```
|-- Medication and treatment-effectiveness logging
|-- Voice or typed note input
|-- Migraine timeline
|-- Migraine Twin
|-- Share with GP
```

Supabase backend

```
|-- profiles
|-- patients
|-- gp_profiles
|-- medications
|-- migraine_events
|-- migraine_symptoms
|-- migraine_triggers
|-- medication_logs
|-- voice_notes
|-- gp_access
|-- reports
|-- appointments
|-- wearable_events
```

GP dashboard on website

```
|-- Patient overview
|-- Key migraine changes
|-- Frequency, severity and duration charts
|-- Medication effectiveness
|-- Trigger context
|-- Migraine Twin
|-- Consultation Copilot
|-- Ask Cerevia
|-- Previous appointment comparison
|-- GP summary report
```

5. Recommended Repository Structure

```
cerevia-demo/
apps/
  website/           # Next.js website and GP dashboard
  ios-patient/       # SwiftUI iOS app project
packages/
  shared-types/      # Optional TypeScript interfaces and JSON schemas
  seed-data/         # Sarah seed data in JSON
  ai-prompts/        # Summary, voice, Ask Cerevia and comparison prompts
supabase/
  migrations/
  seed.sql
  policies.sql
docs/
  architecture.md
  demo-script.md
  safety-boundaries.md
  README.md
```

6. Website Architecture

The website should not become a full patient web app. The patient experience is shown through the iOS app. The website is the public demo entry point and the GP dashboard.

6.1 Website routes

```

/
  Landing page: "Cerevia Hackathon Demo" with Patient and GP choices
/patient
  Patient demo instructions or launch explanation
/gp
  GP dashboard home
/gp/patient/sarah
  Sarah overview
/gp/patient/sarah/timeline
  Migraine event timeline
/gp/patient/sarah/migraine-twin
  GP Migraine Twin
/gp/patient/sarah/consultation
  Consultation Copilot
/gp/patient/sarah/ask
  Ask Cerevia
/gp/patient/sarah/report
  Generated migraine consultation summary
/tech
  Architecture overview for judges
/demo-story
  Explains Sarah's demo journey

```

6.2 Landing page requirements

- Must say: Cerevia Hackathon Demo.
- Must have two clear options: Continue as Patient and Continue as GP.
- The Patient option should explain that the patient experience is shown in the iOS app through Xcode/iOS simulator.
- The GP option should route directly to the GP dashboard.
- Include a short one-line explanation: Cerevia helps GPs understand months of migraine history in minutes.
- Keep the page simple so judges instantly understand what to click.

7. Patient iOS App Architecture

The patient app is the patient-facing product. It should feel like something a real migraine patient would use after attacks and before GP appointments. Since the aim is to show it through Xcode, SwiftUI is the cleanest route unless the team has a strong React Native reason.

Layer	Recommendation	Reason
UI	SwiftUI	Fast native app development with clean previews and iOS simulator support
Backend client	Supabase Swift	Connects directly to Supabase data and auth-lite flows
3D/visual	SceneKit or RealityKit	Suitable for a simple Migraine Twin head model
State	ObservableObject / lightweight local state	Enough for a hackathon app
Voice	Typed transcript first, optional AVFoundation later	Do not block core flow on audio permissions
Charts	Swift Charts if needed	Useful for patient-facing frequency/severity views

7.1 Patient app screens

- Welcome/demo patient screen
- Patient migraine dashboard
- Migraine episode logging screen
- Pain location selection
- Trigger logging screen

- Medication and effectiveness tracking
- Voice or typed note input screen
- Migraine event timeline
- Migraine Twin
- Share with GP
- Appointment preparation

7.2 Patient app core journey

1. Patient opens the app and sees Sarah's migraine dashboard.
2. Patient logs an advanced migraine episode: severity, duration, pain location, symptoms, triggers and impact.
3. The app writes a `migraine_episode_logged` event to Supabase.
4. The patient timeline updates.
5. The Migraine Twin highlights affected regions such as right temple or behind left eye.
6. The patient logs medication taken and perceived effectiveness.
7. The patient adds a voice note or typed transcript.
8. The patient shares access with the GP.
9. The GP dashboard reflects the patient data.

8. GP Dashboard Architecture

The GP dashboard is the clinician intelligence layer. Its job is to answer: has the patient improved, has the patient worsened, is the medication helping, and what should the GP discuss?

8.1 GP dashboard sections

- Patient summary card: name, age, condition, medication, access status and current concern.
- Key migraine changes since last appointment.
- Migraine frequency trend.
- Severity and duration trends.
- Medication effectiveness story.
- Trigger context panel.
- Wearable context panel: sleep, activity and heart rate.
- Migraine event timeline.
- Migraine Twin.
- Consultation Copilot.
- Ask Cerevia.
- Generated GP report.

8.2 GP dashboard success criteria

- A GP should understand Sarah's migraine pattern in under 60 seconds.
- The dashboard should prioritise changes, not raw diary entries.
- Medication effectiveness should be visible as a story, not just a percentage.
- Trigger information should be presented as context and possible patterns, not certainty.
- AI outputs should use safe clinical language.
- The GP should never need to scroll through dozens of migraine entries to understand the current story.
- The Migraine Twin should support understanding, not distract from the consultation.

9. Supabase Backend

Supabase is the shared source of truth between the iOS app and website. Use it for database, storage, auth-lite flows and access rules. For the hackathon, focus on a reliable schema and clear demo behaviour rather than perfect production authentication.

Table	Purpose
profiles	Stores user identity and role: patient, GP or future carer
patients	Stores patient-specific profile information
gp_profiles	Stores GP or practice details
medications	Stores current migraine medication and relevant patient-reported medication context
migraine_events	Main timeline table for migraine episodes and important related events

migraine_symptoms	Stores detailed symptom layers for episodes: aura, nausea, light sensitivity, etc.
migraine_triggers	Stores suspected triggers linked to migraine events
medication_logs	Stores medication taken, time to relief and perceived effectiveness
voice_notes	Stores voice transcript metadata and structured output
gp_access	Controls whether a GP can access a patient's data
reports	Stores generated migraine summaries and appointment packs
appointments	Stores current and previous appointment context
wearable_events	Stores simulated Apple Watch data

9.1 Suggested SQL schema

```
create table profiles (
  id uuid primary key default gen_random_uuid(),
  full_name text not null,
  email text unique,
  role text check (role in ('patient', 'gp', 'carer')) not null,
  created_at timestamptz default now()
);
```

```
create table patients (
  id uuid primary key default gen_random_uuid(),
  profile_id uuid references profiles(id),
  date_of_birth date,
  primary_language text default 'en',
  diagnosis text default 'Chronic migraine',
  created_at timestamptz default now()
);
```

```
create table gp_profiles (
  id uuid primary key default gen_random_uuid(),
  profile_id uuid references profiles(id),
  practice_name text,
  role_title text default 'GP',
  created_at timestamptz default now()
);
```

```
create table medications (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  name text not null,
  dosage text,
  usage_type text, -- acute, preventive, pain_relief, other
  active boolean default true,
  created_at timestamptz default now()
);
```

```
create table migraine_events (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  event_type text not null,
  started_at timestamptz,
  ended_at timestamptz,
  severity int check (severity between 1 and 10),
  duration_minutes int,
```

```

primary_pain_region text,
title text not null,
description text,
impact_score int check (impact_score between 1 and 10),
source text default 'patient_app',
metadata jsonb default '{}'::jsonb,
created_at timestamptz default now()
);

create table migraine_symptoms (
  id uuid primary key default gen_random_uuid(),
  migraine_event_id uuid references migraine_events(id),
  aura boolean default false,
  light_sensitivity boolean default false,
  sound_sensitivity boolean default false,
  nausea boolean default false,
  dizziness boolean default false,
  visual_disturbance boolean default false,
  neck_tension boolean default false,
  notes text,
  created_at timestamptz default now()
);

create table migraine_triggers (
  id uuid primary key default gen_random_uuid(),
  migraine_event_id uuid references migraine_events(id),
  trigger_type text not null,
  trigger_value text,
  confidence text check (confidence in ('suspected', 'likely', 'unknown')) default 'suspected',
  created_at timestamptz default now()
);

create table medication_logs (
  id uuid primary key default gen_random_uuid(),
  migraine_event_id uuid references migraine_events(id),
  medication_id uuid references medications(id),
  taken_at timestamptz,
  effectiveness_score int check (effectiveness_score between 1 and 10),
  time_to_relief_minutes int,
  side_effects text,
  notes text,
  created_at timestamptz default now()
);

create table voice_notes (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  transcript text,
  structured_output jsonb default '{}'::jsonb,
  linked_event_id uuid references migraine_events(id),
  created_at timestamptz default now()
);

create table gp_access (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  gp_id uuid references gp_profiles(id),

```

```

access_status text check (access_status in ('active', 'revoked', 'expired')) default 'active',
access_code text,
starts_at timestamptz default now(),
expires_at timestamptz,
created_at timestamptz default now()
);

```

```

create table reports (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  gp_id uuid references gp_profiles(id),
  report_type text,
  title text,
  summary text,
  structured_output jsonb default '{}'::jsonb,
  date_range_start date,
  date_range_end date,
  created_at timestamptz default now()
);

```

```

create table appointments (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  gp_id uuid references gp_profiles(id),
  appointment_date timestamptz,
  appointment_type text,
  notes text,
  created_at timestamptz default now()
);

```

```

create table wearable_events (
  id uuid primary key default gen_random_uuid(),
  patient_id uuid references patients(id),
  source_device text default 'Simulated Apple Watch',
  timestamp timestamptz default now(),
  sleep_hours numeric,
  steps int,
  resting_heart_rate int,
  active_minutes int,
  metadata jsonb default '{}'::jsonb,
  linked_event_id uuid references migraine_events(id),
  created_at timestamptz default now()
);

```

10. Migraine Event Architecture

The migraine event table is the core of the system. It turns different types of patient activity into one migraine-focused timeline. This powers summaries, charts, timeline replay, Ask Cerevia and the Migraine Twin.

Rule: If it helps the GP understand migraine progression, medication response, trigger context or patient impact, make it a Migraine Event.

Event type	Meaning	Example
migraine_episode_logged	Patient logs a migraine attack	Right temple pain, severity 8, lasted 6 hours
trigger_logged	Patient records possible trigger context	Slept 4.5 hours, high stress, skipped lunch
medication_taken	Patient records medication taken during attack	Sumatriptan taken 45 minutes after onset
medication_effectiveness_logged	Patient records whether treatment helped	Relief after 90 minutes, effectiveness 7/10

voice_note_added	Patient records or types a note	Severe pain behind right eye with nausea
wearable_data_synced	Apple Watch simulation data added	Sleep 4.8h, steps 2,100, resting HR 88
gp_access_granted	Patient shares data with GP	Access active for 7 days
report_generated	GP summary generated	90-day migraine consultation report
appointment_prepared	Appointment pack created	Patient priorities listed

10.1 Example Migraine Event

```
{
  "event_type": "migraine_episode_logged",
  "patient_id": "sarah_patient_uuid",
  "started_at": "2026-06-05T08:20:00Z",
  "ended_at": "2026-06-05T14:40:00Z",
  "title": "Severe migraine behind right eye",
  "description": "Started after poor sleep and high screen time. Light sensitivity and nausea reported.",
  "severity": 8,
  "duration_minutes": 380,
  "primary_pain_region": "behind_right_eye",
  "impact_score": 8,
  "source": "patient_app",
  "metadata": {
    "triggers": ["poor_sleep", "screen_time", "skipped_meal"],
    "symptoms": ["aura", "light_sensitivity", "nausea"],
    "medication_taken": "Sumatriptan",
    "time_to_relief_minutes": 90,
    "work_impact": "missed morning meetings"
  }
}
```

11. Seeded Demo Patient

Use a single deterministic seeded patient. This avoids unreliable demos and allows the team to script a compelling story. Every logged event should support the narrative that the GP can quickly understand changes over time.

Field	Value
Name	Sarah Ahmed
Age	29
Primary language	English
Condition	Chronic migraine
Medication	Sumatriptan
Recent symptoms	Right temple pain, behind-eye pain, aura, light sensitivity, nausea, dizziness, neck tension
Trigger context	Poor sleep, high stress, screen time, skipped meals, dehydration, weather change, menstrual-cycle context
Wearable context	Reduced sleep, lower activity on migraine days, elevated resting heart rate around severe episodes
Appointment context	Upcoming GP appointment with previous appointment comparison
Patient concern	Migraines are becoming more frequent and affecting work productivity

11.1 Suggested 12-week demo narrative

Period	Story beat	Data to seed
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Weeks 1-2	Baseline migraine pattern	2-3 migraines, average severity 5, normal sleep
Weeks 3-4	Sleep starts declining	Sleep drops below 6 hours on several nights
Weeks 5-6	Frequency increases	4-5 migraines, more right-temple pain and light sensitivity
Weeks 7-8	Medication review begins	Sumatriptan used more consistently, variable relief
Weeks 9-10	Impact becomes clearer	Missed work/study, reduced activity on migraine days
Weeks 11-12	Current concern	Severe behind-eye migraines, aura and nausea, GP appointment generated

12. Patient App Detailed Build

12.1 Dashboard

- Greeting and patient name.
- Current month migraine count.
- Average severity card.
- Average duration card.
- Most common pain region.
- Top suspected triggers.
- Medication effectiveness card.
- Upcoming GP appointment card.
- Quick actions: Log Migraine, Add Trigger, Medication, Voice Note, Share with GP.
- Small Migraine Twin preview.

12.2 Advanced migraine episode logging

- Fields: severity, start time, end time, duration, pain region, symptoms, suspected triggers, medication, time to relief, side effects, impact and notes.
- Symptoms should include aura, light sensitivity, sound sensitivity, nausea, dizziness, visual disturbance and neck tension.
- Pain regions should include left temple, right temple, forehead, behind left eye, behind right eye, back of head and neck.
- Impact should capture missed work, reduced productivity, cancelled plans, missed exercise or difficulty studying.
- On submit, create a `migraine_episode_logged` event and linked symptom/trigger/medication rows.
- Update timeline and Migraine Twin immediately.
- Use accessible language and large touch targets.

12.3 Trigger logging

- Triggers are a major part of the platform.
- Allow the patient to record sleep, stress, hydration, caffeine, missed meals, screen time, weather, menstrual-cycle context and other notes.
- Trigger language must remain cautious: suspected trigger, context, possible pattern.
- Trigger logging can be attached to a migraine event or logged as daily context.

12.4 Medication and effectiveness tracking

- Show current medication list.
- Allow patient to record medication taken during a migraine episode.
- Capture time taken after onset.
- Capture time to relief.
- Capture perceived effectiveness from 1-10.
- Capture side effects or no relief.
- Create `medication_taken` and `medication_effectiveness_logged` events.

12.5 Voice or typed note input

- Start with typed transcript input for reliability.
- Optional: add actual audio recording after the core flow works.
- Use AI mock to structure the transcript into migraine fields.
- Create voice_note_added and linked migraine_episode_logged event where appropriate.

12.6 Share with GP

- Patient clicks Share with GP.
- Create active gp_access row.
- Show access code such as CER-4821 or QR code.
- Show access expiry, for example 7 days.
- Include revoke access option, even if simulated.

13. GP Dashboard Detailed Build

13.1 Patient overview

- Sarah's name, age, condition and medication.
- Access status and expiry.
- Top current concerns.
- Latest AI-generated migraine summary.
- Current month migraine count, average severity and average duration.

13.2 Key migraine changes

- New or more frequent symptoms since last appointment.
- Frequency changes.
- Severity changes.
- Duration changes.
- Medication effectiveness changes.
- Trigger context changes.
- Wearable context changes such as sleep, activity and heart rate.

13.3 Charts

- Migraine frequency by week.
- Severity over time.
- Duration over time.
- Medication effectiveness over time.
- Sleep hours against migraine severity.
- Trigger frequency and co-occurrence.
- Impact score over time.

13.4 Timeline

- Chronological list of migraine events.
- Filters by migraine episode, trigger, medication, voice, wearable and report events.
- Highlight newly added events from patient app.
- Group by week for timeline replay.

14. Migraine Twin

The Migraine Twin should be useful before it is impressive. It should not attempt medical diagnosis or detailed anatomy. It should be a simplified 3D head model or clean head-region visual that maps migraine pain regions and symptoms to visual zones.

Experience	Purpose	What to show
Patient app	Help Sarah understand what she has logged	Affected regions, recent migraine episodes, simple severity indicators
GP dashboard	Help GP understand migraine distribution quickly	Most affected regions, new regions, worsening regions, timeline replay overlays

14.1 Head region mapping

left_temple -> left-sided migraine pain
 right_temple -> right-sided migraine pain
 forehead -> frontal pain or pressure
 behind_left_eye -> retro-orbital pain, visual symptoms
 behind_right_eye -> retro-orbital pain, visual symptoms
 back_of_head -> occipital pain
 neck -> neck tension or migraine-associated neck pain
 whole_head -> generalised severe migraine
 visual_layer -> aura or visual disturbance
 sensory_layer -> light sensitivity and sound sensitivity
 nausea_layer -> nausea or vomiting context

14.2 Migraine Twin logic

- Calculate frequency of migraine events per pain region.
- Calculate average severity per pain region.
- Increase intensity for recent or repeated migraine regions.
- Overlay aura, light sensitivity and nausea as simple badges or rings.
- Show a legend so judges understand what the visual means.
- During timeline replay, update the Migraine Twin as each period is shown.

15. Simulated Apple Watch Feature

For the hackathon, build this as a simulated data flow rather than real HealthKit integration. The goal is to show how objective context can sit alongside patient-reported migraine events.

Simulated Apple Watch data
 -> Sleep: 4.8 hours
 -> Steps: 2,100
 -> Resting heart rate: 88 bpm
 -> Active minutes: 14

Converted into:
 -> wearable_data_synced event
 -> wearable_events row
 -> GP wearable context panel
 -> Migraine Twin overlay
 -> AI summary context

- Do not claim diagnosis from wearable data.
- Use wording such as context, pattern, correlation and may be worth discussing.
- Show relationships like poor sleep appearing near severe migraine episodes, but do not infer disease.
- Sleep, activity and heart rate should be visible because the user selected Apple Watch focus: sleep + activity + heart rate.

16. AI Layer

The AI layer should be built as a set of separate functions. Mock all outputs first. Once the app and dashboard are working, replace mocks with live API calls if time allows.

Function	Input	Output
Migraine consultation summary	Migraine events over date range	200-word GP-ready migraine summary
Voice note structuring	Transcript	Structured migraine episode fields
Ask Cerevia	GP question plus migraine timeline	Grounded answer from patient history
Pattern detection	Events, triggers and wearable data	Safe trend flags
Previous appointment comparison	Two date ranges	New, worsened and improved migraine items
Medication effectiveness	Medication logs and episodes	Patient-reported response summary
Trigger context	Triggers linked to migraine events	Commonly reported context before severe

		episodes
--	--	----------

16.1 Migraine consultation summary prompt

You are preparing a GP consultation summary from patient-reported migraine events.
Do not diagnose, prescribe, recommend medication changes or give treatment advice.
Use neutral language such as "reported", "appears", "patient-reported", "may be worth discussing" and "suggested consultation focus".

Return:

1. Brief overview
2. Migraine frequency, severity and duration trends
3. Pain-location and associated symptom patterns
4. Medication use and patient-reported effectiveness
5. Trigger context and wearable context if relevant
6. Patient priorities and daily impact
7. Suggested GP discussion areas
8. Safety disclaimer

Patient migraine events:

```
{{migraine_events}}
```

16.2 Ask Cerevia example questions

- Has migraine frequency worsened since the previous appointment?
- Which pain region is reported most often?
- How effective was Sumatriptan over the last month?
- What symptoms are becoming more common?
- What context appears near severe migraine episodes?
- How often did poor sleep appear before severe attacks?
- What is Sarah most worried about?
- What changed after the last appointment?

17. Trigger Intelligence

Triggers are a major part of Cerevia. However, the platform must avoid claiming causation. It should present trigger information as patient-reported context and possible patterns that may be useful for discussion.

Trigger category	Examples	How to present
Sleep	Sleep hours, poor sleep, irregular sleep	Common context before severe episodes
Stress	Work stress, exam stress, anxiety	Patient-reported high-stress days
Hydration	Low water intake, dehydration	Possible context logged before episodes
Food/routine	Skipped meals, caffeine changes	Reported lifestyle context
Screen time	Long laptop/phone use	Reported before some episodes
Weather	Weather change, pressure change	Patient-suspected context
Menstrual-cycle context	Cycle timing where relevant	Private patient-reported context for discussion

18. Consultation Copilot

Consultation Copilot is the GP's one-click preparation feature. If you only had one advanced GP feature, this would be the most useful.

Input:

```
Patient profile
Recent migraine events
Previous appointment
Medication logs
Trigger context
```


Patient priorities
Wearable context

Output:

What changed
What matters most
Whether frequency, severity or duration worsened
Whether medication appears to be helping based on patient-reported data
What the patient is worried about
Suggested discussion areas
Questions the GP may want to ask
Short safety disclaimer

18.1 Example Consultation Copilot output

Sarah has reported an increase in migraine frequency over the last 8 weeks, from 2-3 episodes per month to 8 episodes this month. Average severity has increased from 5.1 to 7.3, and recent episodes more often involve right-temple or behind-right-eye pain with aura and nausea.

Sumatriptan was used in 6 recent episodes. Patient-reported relief was variable, with time to relief ranging from 60 to 150 minutes.

Commonly reported context before severe episodes includes poor sleep, high screen time and skipped meals. Wearable data also shows reduced sleep before several severe episodes.

Suggested consultation focus:

- Review increasing migraine frequency and impact on work.
- Discuss medication response and timing.
- Ask about aura, nausea and behind-eye pain pattern.
- Review sleep and lifestyle context as discussion points.

This summary is based on patient-reported data and does not provide diagnosis or treatment advice.

19. Previous Appointment Comparison

This is highly useful for doctors because it focuses on change, not raw history.

Since last appointment:

New or more common symptoms:

- Aura reported in 4 recent episodes
- Nausea reported more frequently
- Behind-right-eye pain reported for the first time this period

Worsened:

- Migraine frequency increased from 3 per month to 8 this month
- Average severity increased from 5.1 to 7.3
- Average duration increased from 3.4 hours to 5.8 hours

Medication:

- Sumatriptan used more consistently
- Patient-reported relief remains variable

Trigger/wearable context:

- Poor sleep appears near several severe episodes
- Lower activity recorded on migraine days

Suggested focus:

- Discuss increasing frequency and work impact
- Review medication response and timing

- Ask about aura and nausea pattern

20. Timeline Replay

Timeline Replay is the visual story of the patient's migraine journey. It should be useful for the GP, not just animated for effect.

Week 1: Two migraines, average severity 5, no aura
 Week 2: Poor sleep begins appearing in logs
 Week 3: Right-temple pain becomes more common
 Week 4: Frequency increases to three attacks in one week
 Week 5: Sumatriptan used in multiple episodes
 Week 6: Behind-eye pain and nausea appear
 Week 7: Work impact increases
 Week 8: GP summary generated before appointment

- Use events grouped by week or month.
- Update the Migraine Twin as the replay moves forward.
- Show key turning points rather than every single log.
- End with a summary of what changed.

21. Safety and Clinical Boundaries

Non-negotiable safety rule: Cerevia does not diagnose, prescribe, recommend treatment changes or replace a clinician. It structures patient-reported migraine information to support a GP consultation.

Unsafe wording	Safer wording
Your migraine is caused by poor sleep	Poor sleep was frequently reported before severe episodes and may be worth discussing
Increase your medication	Medication response changed and may be worth reviewing with the GP
This means your treatment is failing	Patient-reported relief was variable during this period
You have a neurological problem	Aura and visual symptoms were reported and may be worth discussing
Diagnosis	Patient-reported summary
Treatment recommendation	Suggested consultation focus

22. Build Order

22.1 Pre-hackathon build

10. Agree the final product story and Sarah demo narrative.
11. Create the Supabase project and schema.
12. Seed Sarah's patient profile, medication, appointments, migraine events, trigger context and wearable data.
13. Create the Next.js website and landing page.
14. Create the SwiftUI iOS app project in Xcode.
15. Connect both projects to Supabase.
16. Build the basic patient dashboard and GP dashboard.
17. Create static mock AI outputs.
18. Build initial Migraine Twin prototypes in both experiences.
19. Prepare reusable pitch wording and demo script.

22.2 Hackathon build

20. Polish the landing page and demo flow.
21. Connect patient actions to live Supabase events.
22. Make the GP dashboard respond to new patient events.
23. Add Consultation Copilot and Ask Cerevia.
24. Add previous appointment comparison.
25. Add timeline replay and Migraine Twin updates.
26. Add simulated Apple Watch sync.
27. Add trigger intelligence and medication effectiveness views.

28. Test the full judge path repeatedly.
29. Prepare final pitch and fallback screenshots.

23. Team Split

Role	Responsibilities
Product/Pitch Lead	Own problem framing, demo story, safety wording, feature prioritisation and final pitch
iOS Patient App Lead	Build SwiftUI patient dashboard, migraine logging, timeline, sharing and Migraine Twin
Website/GP Dashboard Lead	Build landing page, GP dashboard, charts, Consultation Copilot and Ask Cerevia UI
Backend/Supabase Lead	Own schema, seed data, access logic, Supabase clients and data reliability
AI/Intelligence Lead	Own summaries, voice structuring, Ask Cerevia, comparison logic and trigger context outputs
Visual/Migraine Twin Lead	Own 3D/visual model, head region mapping, timeline replay and wearable overlays

24. Acceptance Criteria

- Website landing page says Cerevia Hackathon Demo.
- Landing page offers Patient and GP choices.
- Patient experience opens in Xcode/iOS simulator.
- Patient can log at least one advanced migraine episode, one trigger context entry, one medication-effectiveness event and one voice note/transcript.
- Those actions create Migraine Events in Supabase.
- GP dashboard reads Sarah's timeline from Supabase.
- Migraine Twin appears in both patient and GP experiences.
- GP can view key changes and previous appointment comparison.
- GP can use Consultation Copilot.
- GP can ask Cerevia at least three questions about Sarah's migraine history.
- Wearable simulation appears in timeline and GP context.
- Trigger context and medication effectiveness are visible.
- AI output uses safe language and avoids diagnosis.
- Demo works without relying on unpredictable random data.

25. Demo Flow for Judges

30. Open the website landing page and show the two routes: Patient and GP.
31. Open the patient app in Xcode/iOS simulator.
32. Show Sarah's dashboard and explain her migraine background.
33. Log a severe migraine episode with right-temple or behind-right-eye pain.
34. Add aura, light sensitivity, nausea and suspected triggers.
35. Record Sumatriptan use and perceived effectiveness.
36. Show the Migraine Event created and timeline updated.
37. Show the Migraine Twin updating in the patient app.
38. Log a voice note transcript.
39. Share access with the GP.
40. Switch to the GP dashboard on the website.
41. Show frequency, severity, duration, trigger context, medication effectiveness, Migraine Twin and timeline.
42. Run Consultation Copilot.
43. Ask Cerevia: Has migraine frequency worsened since the last appointment?
44. Show previous appointment comparison.
45. Show simulated Apple Watch sleep, activity and heart-rate context.
46. End with the GP-ready migraine summary and safety boundary.

26. Suggested Demo Script

This is Cerevia, an AI-powered migraine management platform built to help GPs understand months of migraine history in minutes.

Our demo patient is Sarah, a 29-year-old chronic migraine patient. Over the last few months, her migraines have become more frequent, more severe and more disruptive to work. Like many patients, she struggles to remember exact frequency, duration, triggers and medication response during a short GP appointment.

In the patient app, Sarah can log detailed migraine episodes, pain location, aura, symptoms, suspected triggers, medication use and daily impact. Each entry becomes a structured Migraine Event, building a timeline of her migraine history between appointments.

The Migraine Twin gives Sarah and her GP a visual view of where pain is happening and how patterns change over time. The GP dashboard then turns Sarah's migraine history into key changes, trends, medication-effectiveness insights, trigger context, wearable context and an AI-generated consultation summary.

Cerevia does not diagnose or prescribe. It organises patient-reported migraine data so the GP can prepare faster, ask better questions and make more informed clinical decisions.

27. Fallback Strategy

If something breaks during the live demo, the team should still be able to show the story. Prepare seeded views, screenshots or static mock outputs for critical features.

If this breaks	Fallback
Live AI	Use pre-generated mock summary
Supabase write	Use seeded Migraine Events already in the database
iOS simulator	Use recorded screen capture of patient flow
Migraine Twin	Use static head visual with severity overlays
Wearable sync	Use pre-seeded Apple Watch data card
Ask Cerevia	Use saved example questions and answers

28. Roadmap After Hackathon

- Real HealthKit integration for Apple Watch data.
- GP system integration in the long term.
- More robust patient consent and access controls.
- More detailed migraine medication and treatment-response tracking.
- Optional neurologist-facing specialist view after GP validation.
- Accessibility testing with migraine patients who experience light sensitivity.
- Clinical feedback sessions with GPs.
- Formal security and privacy review.
- Pilot with GP surgeries after validation.

29. Final Reminder for the Team

Do not pitch Cerevia as a migraine tracker. Pitch it as a migraine consultation-intelligence platform that helps clinicians understand months of migraine history in minutes.

The strongest final product is not the one with the most features. It is the one where every feature supports the same story: patient logs migraine history over time, Cerevia structures the information, the GP understands the pattern faster, and the consultation improves.